

## 2.13 Green chemistry

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### Students will be assessed on their ability to:

- a demonstrate an understanding that the processes in the chemical industry are being reinvented to make them more sustainable ('greener') by:
  - i changing to renewable resources
  - ii finding alternatives to very hazardous chemicals
  - iii discovering catalysts for reactions with higher atom economies, e.g. the development of methods used to produce ethanoic acid based on catalysts of cobalt, rhodium and iridium
  - iv making more efficient use of energy, e.g. the use of microwave energy to heat reactions in the pharmaceutical industry
  - v reducing waste and preventing pollution of the environment
- b discuss the relative effects of different greenhouse gases as absorbers of IR and hence on global warming
- c discuss the difference between anthropogenic and natural climate change over hundreds of thousands of years
- d demonstrate understanding of the terms *carbon neutrality* and *carbon footprint*
- e apply the concept of carbon neutrality to different fuels, such as petrol, bio-ethanol and hydrogen
- f discuss and explain, including the mechanisms for the reactions, the science community's reasons for recommending that CFCs are no longer used due to their damaging effect on the ozone layer.